

# Ethan Winn Bishop

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Provo, UT | Open to Relocation

## WORK EXPERIENCE

**VFX/Technical Artist** - BYU Animation AUGUST 2024 - PRESENT

- Led a team of 3 VFX artists to create realtime VFX and tools for capstone game 'Dragonkisser'
- Used Unreal Widget Blueprints, Blueprints, C++, and Niagara to create environment, FX, and UI tools within Unreal and Houdini to allow artist control while maintaining instances and shader optimizations
- Worked in a team of 4 FX artists to develop FX and rendering tools in Houdini for capstone short film 'Honey Business' including honey, grass, bee swarms, and USD tools for instanced variants

**Teaching/Research Assistant** - BYU Animation AUGUST 2024 - MAY 2025

- Assisted 3 classes of 20-30 students in understanding programming, procedural modeling, rendering, FX, and shading in Houdini and Maya through one-on-one and group sessions
- Worked in a team of 8 FX artists to develop FX tools within Houdini for capstone short film 'Love & Gold' including torch fires, dragon fire blasts, coin RBD, and various magic FX

**Motion Graphics Technical Artist** - BYU Broadcasting SEPTEMBER 2022 - JANUARY 2025

- Trained 6 new hires in C4D, Blender, After Effects
- Scripted procedural animation and template tools within C4D, Blender, Houdini, and After Effects
- Assisted setting up a render farm network using Deadline

## PROJECTS

**Realtime GPU Pathtracer**

- Used the Vulkan API within a C++ framework to create a realtime pathtracer capable of 60 FPS at 1080p with over 250,000 triangles on RTX 4070 Ti hardware with animated lights and geometry
- Create the shaders in GLSL (raygen, PBR materials, cook-torrence BRDF, and denoising)
- Scripted a bash tool to automatically convert shaders to slang at build time

**Digging Mini Game**

- Created a custom Skidsteer vehicle in UE5 and C++ that can add and remove to/from a custom C++ voxel terrain class
- Coded a custom script in Houdini that prepares and exports VDBs for use in voxel terrain generation
- Used UE5, C++, and Houdini to create a custom vehicle system based on UE5 Chaos Vehicle for specific movement and collision with the unusual terrain

## EDUCATION

**Brigham Young University - Provo, UT**

SEPTEMBER 2021 - APRIL 2026

BS in Computer Science Animation | GPA: 3.89

## SKILLS

**Programming** - C++, C, Python, VEX, GLSL, HLSL, Java, JavaScript, OpenMP

**3D Software** - Houdini, Maya, Cinema4D, Blender (including geometry nodes), EmberGen

**Game Engines** - Unreal Engine (including Material graph, Blueprints, Niagara), Godot, Unity

**Shading & Rendering** - Substance Painter, Substance Designer, RenderMan, Redshift, Arnold

**Tools & Technologies** - OpenGL, Vulkan, Git, Perforce, NVIDIA Nsight, Unreal Insight, CMake, Solaris, USD, Linux

## REFERENCES

Seth Holladay 801-422-6490 holladay@cs.byu.edu | Parris Egbert 801-422-4029 egbert@cs.byu.edu

## CLUBS AND EXTRACURRICULAR ACTIVITIES

**BYU Animation Club**



**BYU CS Group**  
JANUARY 2022



**Club**

MONTH 2022 - MONTH 2000

**Club**

MONTH 2022 - MONTH 2000

**Club**

MONTH 2022 - MONTH 2000

## **140 Word Pitch**

In 6th grade, I started to wonder how video games were made and I started experimenting with creating my own games. I used Scratch and Adobe Flash to make some simple sprite-based games, which at the time I felt were practically ready to be a AAA release (as you can imagine, they weren't quite there yet). As I have developed my skills in video game creation, I have put effort into learning art, design, physics, and computer science in order to blend art and design with realism and technical knowledge, building for myself a strong foundation for creating real time effects and shaders for use in video games.

I use design, physics, and cs to blend artistry with technical knowledge, with a passion and focus on creating real time effects and shaders